IEICE TRANS. COMMUN., VOL.E108–B, NO.2 FEBRUARY 2025

PAPER

Non-Contact Reconstruction of ECG Signals Using FMCW Radar and SS-S2SA Network

Renwei CUI[†], ***Student Member***, **Wei CUI**[††a)], **Yujian CAI**[†], ***and*** **Yu YAN**[†], ***Nonmembers***

SUMMARY The electrocardiogram (ECG) signals P-wave, QRS wave and T-wave all reflect the activity of the heart, and the analysis of ECG signals can provide basic information for the diagnosis and prevention of heart disease. In the work of this paper, frequency-modulated continuous-wave (FMCW) radar and deep learning network are utilized to acquire ECG signals non-contactly, and we propose an improved differential and cross multiply (DACM) algorithm and a multi-neighbor differentiator for extracting cardiac motion acceleration information, as well as a partitioned reconstruction network incorporating an attention mechanism of encoder-decoder to achieve ECG signal reconstruction. The design principle is a combination of signal segmentation and deep learning (Sequence-to-sequence and attention) called SS-S2SA. firstly, a segmentation algorithm is applied to segment the acceleration signal and the ECG signal synchronously, and then the cardiac motion acceleration signal is mapped to the ECG signal using the SS-S2SA network. The method proposed in this paper is demonstrated to reconstruct ECG signals more accurately and finely by training more than 18,000 acceleration signal segments from 10 healthy subjects and evaluating the predictions from 5 subjects. The average correlation coefficient between the predicted signal and the real signal is about 0.92, and the mean absolute error (MAE) of the timing of the P-peak, R-peak, and T-peak are 13.9 ms, 8.1 ms, and 11.1 ms, respectively.

key words: *FMCW radar, electrocardiogram, radar signal processing, vital sign detection*

1. Introduction

In recent years, human heart health has significantly deteriorated due to an increase in unhealthy lifestyles, physical inactivity, and other factors [1]–[3]. Electrocardiogram (ECG) signals are important biomedical signals that reflect the underlying physiological characteristics of the human body and provide a reliable way for the diagnosis and prevention of cardiovascular diseases [4]. Currently, the main ECG signal acquisition methods are very mostly contact devices [5], [6], which detect the small potential changes caused by the electrical activity generated by myocardial contraction in different parts of the body through electrodes attached to the human skin, and there are still some limitations in some application environments and use scenarios. For example, when using a contact ECG collector to obtain ECG signals from highly infectious patients, device surfaces may transmit pathogens through direct contact or contamination, increasing the risk of infection for healthcare workers. When using a contact ECG collector on burned skin, electrode contact can cause significant pain and discomfort and increase the patient's risk of infection due to the lack of protective lining in the skin. With the rapid development of millimeter-wave radar technology in the field of vital signs monitoring [7]–[11], the use of millimeter-wave radar to obtain ECG signals can be applied to application scenarios that cannot be covered by the contact devices mentioned above [12]–[14].

In this section, we describe other existing studies related to this work. In [15], the U. Ha et al. designed a system that combines signal processing with deep learning to successfully reconstruct seismocardiogram (SCG) waveforms, and the timing of five cardiac mechanical activity events can be obtained with high precision from a single heartbeat. Since the process of cardiac activity is accomplished by the functional mechanical movement of the heart [16], [17], at the same time, the mechanical activity of the heart itself will lead to changes in the electrical activity of the heart. Therefore, theoretically the mechanical activity of the heart and the electrical activity of the heart belong to different domain mapping of the same source information [18]. In recent years, deep learning networks have been heavily used to learn the intrinsic laws of sample data and the mathematical relationships between samples [19], which provides the possibility of non-contact ECG signal monitoring. At this stage of research, J. Park et al. proposed a method to reconstruct ECG signals by SCG signals using a deep learning model with a bidirectional long short-term memory (Bi-LSTM) network [20], [21]. K. Yamamoto et al. [22], [23] proposed to use in-phase (I) and quadrature-phase (Q) signals in the frequency range of 8–30 Hz as neural network inputs to reconstruct ECG signals by a hybrid convolutional neural network (CNN) and long short-term memory (LSTM) deep learning model to reconstruct ECG signals, and the results showed that the mean absolute error (MAE) of this method for P, R and T peaks were 28 ms, 17 ms and 30 ms, respectively. However, the constant change of phase with distance results in no significant features of the phase, which requires high performance of the reconstruction network and is not favorable for porting to run on embedded devices. D. Toda et al. [24] proposed a method for reconstructing ECG signals using CNN network by using frequency-modulated continuous-wave (FMCW) radar to acquire chest displacement signals and performing differential operations as inputs to the neural network. The results confirmed that the method was able to

Manuscript received April 9, 2024.
Manuscript revised June 30, 2024.
Manuscript publicized August 30, 2024.

[†]School of Electronic Information Engineering, Changchun University of Science and Technology, Changchun, JL431, China.
[††]Jilin Province Medical Intelligent Technology and Equipment Joint Technology Innovation Laboratory, Changchun, JL431, China.
a) E-mail: cuiwei@cust.edu.cn (Corresponding author)

DOI: 10.23919/transcom.2024EBP3061

Copyright © 2025 The Institute of Electronics, Information and Communication Engineers

reconstruct ECG signals with a correlation coefficient value of 0.86 and an MAE of 102.2 ms for R-R interval (RRI). J. Chen et al. [25] designed a hybrid deep learning network to convert radio frequency (RF) signals into ECG signals and output them. The reconstructed Q, R, S, and T wave timing errors of this method were 14 ms, 3 ms, 8 ms, and 10 ms. All of the above methods directly use long time series data as the input to the neural network, and long time series data introduces more features and more complex mapping relationships, which makes it more difficult for the model to learn and generalize efficiently in high-dimensional space, which leads to unsatisfactory reconstruction results.

In this paper, we propose a method (i) to use FMCW radar to acquire cardiac motion acceleration signals in the range of 0.9–50 Hz as input to the neural network. (ii) A new encoder-decoder partition reconstruction network with a fusion attention mechanism is designed, and its design principle is a combination of signal segmentation and deep learning (Sequence-to-sequence and attention), called SS-S2SA. This method can accurately reconstruct ECG signals. First we extract the phase signals of thoracic vibration using a modified differential and cross multiply (DACM) algorithm. Then the acceleration signal corresponding to the radar distance signal is directly extracted by the multi-neighborhood difference algorithm to suppress the respiratory motion and amplify the heartbeat motion, which improves the signal-to-noise ratio of the input signal to the neural network and further improves the reconstruction performance. Then, the peak-point-based signal segmentation algorithm is used to synchronize the segmentation process of ECG and acceleration signals, and then these two sets of signals are used to train the SS-S2SA network to map the acceleration signal segments to ECG signal segments, which avoids more features introduced by long sequential data and allows the model to learn the mapping relationship easily in high-dimensional space. Finally, the ECG signal fragments are spliced in chronological order to obtain the complete ECG signal. In order to verify the effectiveness of the method, this study conducted experiments on ECG signal reconstruction based on FMCW radar and SS-S2SA network. The experimental results show that the ECG signal reconstructed by this method is of high quality, and the peak timing error is within the allowable range. The main contributions of this paper are as follows:

1. In this paper, we propose to use FMCW radar to acquire cardiac motion acceleration signals in the range of 0.9–50 Hz as the input to the neural network, which improves the signal-to-noise ratio of the input signal to the neural network and further improves the reconstruction performance.
2. In this paper, a new encoder-decoder partitioned reconstruction network incorporating the attention mechanism, called SS-S2SA, is designed. a more refined reconstruction of each band of the ECG signal is achieved, which improves the reconstruction performance.
3. The method is able to reconstruct the ECG signal waveforms better, and the timing errors of its P-peak, R-peak, and T-peak are within the permissible range. In addition, the reconstruction performance was evaluated for different research subjects.

The remaining parts of this paper are organized and laid out as follows: the second section describes the principle of the methodology for ECG signal reconstruction using FMCW radar, the third section describes the experimental environment and configuration, the fourth section describes the experimental procedure and the analysis of the results, and finally the conclusion is given in the fifth section.

2. Electrocardiographic Reconstruction Method

The ECG signal reconstruction process is shown in Fig. 1, where the radar echo signal is first preprocessed to obtain the chest vibration phase signal; then the cardiac motion acceleration signal is extracted, and the acceleration signal and the ECG signal are synchronously segmented to obtain the corresponding signal segments; finally, the ECG signal is reconstructed by the trained SS-S2SA network model. Next, we will introduce the detailed information of each part later.

2.1 Radar Signal Preprocessing

In this paper, the selected FMCW radar transmit waveform is a sawtooth waveform, and the working block diagram of signal transmission and receiving echo is shown in Fig. 2.

According to [26], the transmit signal of the FMCW radar Tx can be expressed as:

$$S_T(t) = A_T \cos\left(2\pi\left(f_c t + \frac{1}{2}\mu t^2\right) + \varphi_0\right) \tag{1}$$

Where A_T is the transmit signal amplitude, f_c is the carrier frequency, $\mu = B/T_c$ is the frequency modulation (FM) slope, B is the bandwidth, T_c is the width of the FM pulse, and φ_0 is the initial phase of the transmit signal.

Assuming that $R(t)$ is the motion displacement of the

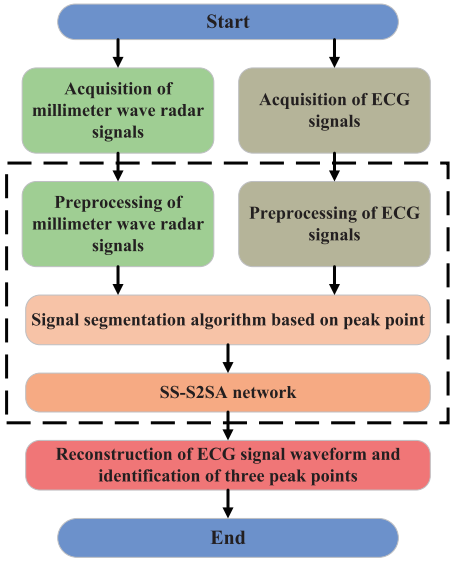


Fig. 1 The overall process of ECG reconstruction.

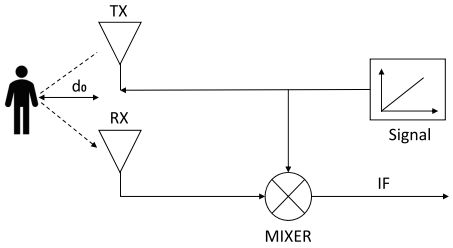


Fig. 2 Principle diagram of IF signal formation.

chest and d_0 is the distance of the target from the radar sensor, the distance from the target's chest to the radar is $x(t) = R(t) + d_0$, the time delay is $t_\tau = 2x(t)/c$, and c is the speed of light. Then the received echo signal can be expressed as:

$$S_R(t) = A_R \cos\left(2\pi\left(f_c\,(t-t_\tau) + \frac{1}{2}\mu\,(t-t_\tau)^2\right) + \varphi_0 + \Delta\varphi\right) \quad (2)$$

Where A_R is the received signal amplitude. The transmitted signal is mixed with the received signal through the mixer to obtain the intermediate frequency signal (IF):

$$S_{IF}(t) = A_T A_R \cos\left(2\pi f_b t + \phi(t)\right) \quad (3)$$

$$f_b = \frac{2Bd_0}{cT_c} \quad (4)$$

$$\phi(t) = \frac{4\pi x(t)}{\lambda} \quad (5)$$

Equations (4) and (5) show that at a fixed distance of d_0, $\phi(t)$ changes with respect to λ as $R(t)$ changes. That is, a target at a fixed distance to move a small distance ΔR, then the phase change between two consecutive times at the target distance unit is:

$$\Delta\phi = \frac{4\pi}{\lambda}\Delta R \quad (6)$$

A distance-fast fourier transformation (FFT) transform of the IF signal will yield the distance information of all reflected signals from the environment, firstly, the interference of static clutter around the target is removed by the moving target detection (MTD) algorithm [27], next multiple distance units are obtained by distance-FFT and the measured target is separated by calculating the energy of each distance unit. Finally, the improved DACM algorithm is utilized to extract the true phase signal of the target. The principle of the traditional DACM algorithm is an approximate realization of a discrete form of the signal inverse tangent differentiation followed by integration [28]. The IF signal after distance FFT can be expressed as:

$$S(t) = I(t) + jQ(t) \quad (7)$$

Where $I(t)$ denotes the real part of the signal and $Q(t)$ denotes the imaginary part of the signal. Firstly, the inverse tangent function is converted to derivative operation, i.e.:

$$\frac{d}{dt}[\phi(t)] = \frac{d}{dt}\left[\arctan\frac{Q(t)}{I(t)}\right] = \frac{I(t)Q'(t) - I'(t)Q(t)}{I(t)^2 + Q(t)^2} \quad (8)$$

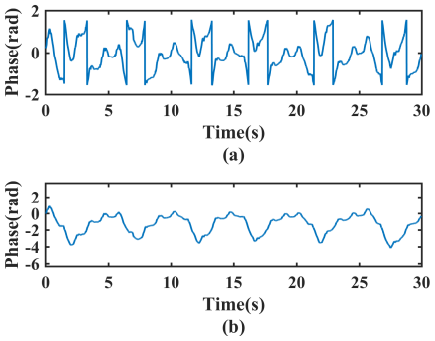


Fig. 3 Results of different phase extraction methods. (a) Direct arctangent function. (b) Improved DACM algorithm.

Where, $I'(t)$ and $Q'(t)$ are the differential forms of $I(t)$ and $Q(t)$ respectively, as the differentiation will make the high frequency noise interference enhanced [29]. In order to suppress the noise, this paper utilizes the nearby data points to offset the effect of noise, in a known $I[k]$ and $Q[k]$ a series of data points (x_k, y_k), $k = 1, 2, 3, \ldots, n$, with equally spaced horizontal coordinates and the value of 1, we can use the five known data points to interpolate to obtain a unity quadratic polynomial, and after calculating the coefficients of the polynomial, we can get the fourth-order differential expression for the first-order derivatives:

$$I'(x_k) = \frac{I(x_{k-2}) - 8I(x_{k-1}) + 8I(x_{k+1}) - I(x_{k+2})}{12}$$
$$Q'(x_k) = \frac{Q(x_{k-2}) - 8Q(x_{k-1}) + 8Q(x_{k+1}) - Q(x_{k+2})}{12} \quad (9)$$

$I'(t)$ and $Q'(t)$ are calculated by the above equation, and Eq. (9) is substituted into Eq. (8), which can be further expressed as a discrete form, and then the signal is integrated to reduce the high frequency component of the signal for the purpose of denoising. The improved DACM algorithm is as follows:

$$\phi[n] = \sum_{k=3}^{n}\left\{\frac{I[k](Q[k-2] - 8Q[k-1] + 8Q[k+1] - Q[k+2])}{12\,(I[k]^2 + Q[k]^2)} - \frac{(I[k-2] - 8I[k-1] + 8I[k+1] - I[k+2])Q[k]}{12\,(I[k]^2 + Q[k]^2)}\right\} \quad (10)$$

Where n denotes the pulse index. We apply the improved DACM algorithm to directly find out the actual phase information of the target without the phase breakpoint problem, which improves the noise suppression effect. Its comparison with the result of inverse tangent demodulation is shown in Fig. 3.

From Eq. (6) it is known that the mixed motion of human respiration and heartbeat leads to the change of phase. The frequency range of respiratory motion is mainly distributed in the range of 0.1–0.5 Hz, and the frequency range of cardiac motion is mainly distributed in the range of 0.9–2 Hz [30], resulting in frequency components distributed in the range

of 8–50 Hz [31], and most of the phase of the RF signal is dominated by the respiratory motion. Since respiratory motion is usually slow, whereas cardiac motion involves rapid muscle contractions. Therefore, the acceleration of the respiratory motion is smaller than that of the cardiac motion, and the respiratory motion can be suppressed and the cardiac motion can be amplified by directly extracting the acceleration change corresponding to the displacement change [32]. The phase signal is first converted to a distance signal by the following equation:

$$d(t) = \phi(t)\frac{\lambda}{4\pi} \tag{11}$$

Where $\phi(t)$ is the phase value at time t, $d(t)$ is the distance value at time t, and λ is the wavelength of the RF signal. Next, the distance signal is differentiated twice to obtain the acceleration signal, and the differentiation operation can be rewritten as a differential operation in the digital domain. Considering that the RF hardware system will generate certain random high-frequency phase noise during operation, and in the acceleration signal extraction process, the traditional forward or backward differentiation will enhance the interference of high-frequency noise. In order to suppress the above noise, we improve the traditional difference method and propose an acceleration signal extraction algorithm based on multi-neighbor difference, which has the advantage over the traditional difference method in that it can extract the acceleration in the low-frequency band more accurately while suppressing the high-frequency system noise and the differential noise, so that the subsequently trained network model has better robustness. The expression of the acceleration at the point in the improved acceleration extraction algorithm is:

$$a_k = \frac{-d_{k-2} + 16d_{k-1} - 30d_k + 16d_{k+1} - d_{k+2}}{12t_T^2} \tag{12}$$

Where d_k is the distance value of point k in the time series, t_T is the frame period between consecutive samples, and a_k is the acceleration of point k in the time series d_k. It should be noted that, according to Eq. (12), when making a five-point difference, it must be calculated from the third point until the end of the third-to-last point, which avoids boundary problems and ensures that the signal sequence does not take negative values. If not enough points can be taken near the boundary point, the traditional two-point difference algorithm is used. Since the main frequency of cardiac motion as well as the resulting frequency components are distributed above 0.9 Hz, the respiratory and other noise signals are finally filtered out by a 0.9–50 Hz bandpass filter to retain the cardiac motion acceleration signal. In order to visualize the relationship between the mechanical and electrical activities of the heart, in this paper, the cardiac motion acceleration signals and the ECG signals are displayed in the same graph, as shown in Fig. 4.

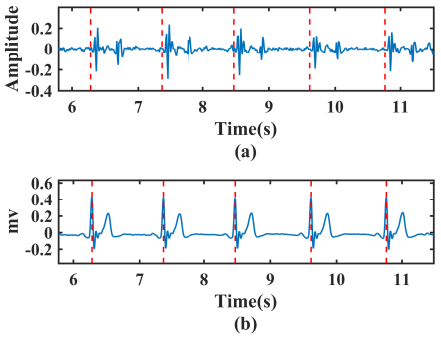


Fig. 4 Cardiac motion acceleration signal extraction results. (a) Cardiac motion acceleration signal. (b) ECG signal.

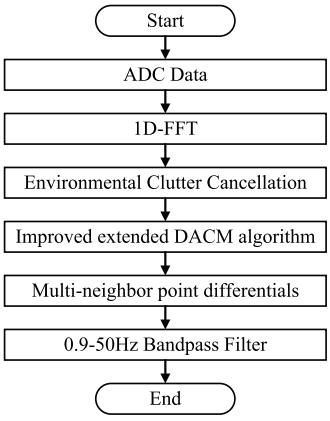


Fig. 5 Radar signal processing flow.

2.2 Peak Point Based Signal Segmentation Algorithm

The ECG signal waveform consists of P wave, QRS complex wave, and T wave. The ECG signal is periodic, and the degree of fluctuation is calculated according to the average amplitude and slope size of the waveform, and the ECG waveform is divided into three regions according to the degree of fluctuation, and the waveforms within the three regions have different characteristics and can be reconstructed separately, which enables a more refined reconstruction of the ECG signal. The single cardiac cycle vector of the ECG signal is denoted as $S_p = [m, m+1, m+2, \ldots, n]$, and the calculation of the characteristics is shown in the following equation:

The average amplitude formula:

$$A_P(m,n) = \frac{\sum_{i=m}^{n} a_p(i)}{n-m+1} \tag{13}$$

Slope formula:

$$K_p(i) = \frac{a_p(i) - a_p(i-1)}{pos(i) - pos(i-1)} \tag{14}$$

Where $a_p(i)$ denotes the amplitude at the i th sampling point of a single cardiac cycle, and $pos(i)$ denotes the i th sampling point. Figure 6 shows the schematic diagram

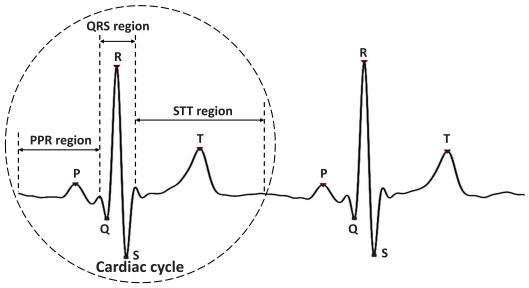


Fig. 6 Schematic diagram of ECG signal waveform partition reconstruction.

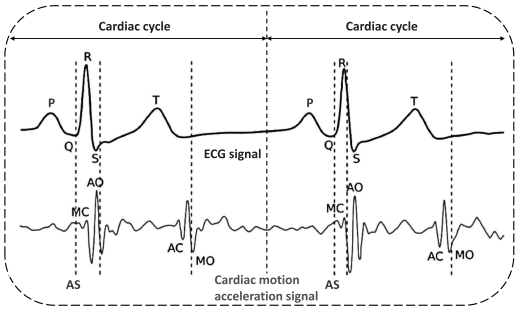


Fig. 7 Schematic diagram of synchronized segmentation of ECG and acceleration signals.

of ECG signal waveform partitioning reconstruction. The specific partitioning method is as follows:

(1) PPR region

The PPR region consists of the P and PR bands, which have smaller average amplitudes and lower slopes, and the fluctuations in this region are smaller.

(2) QRS region

The band consisting of QRS complex wave clusters has the highest average amplitude and the largest slope, and this region has the most intense fluctuations and is the region with the most complex waveforms.

(3) STT region

The STT region consists of the ST band and the T wave, which has a higher average amplitude and a larger slope, and the region fluctuates more violently.

For the cardiac motion acceleration signal, the three peak points AS, AO, and MO are defined as the characteristic points of the acceleration signal, as shown in Fig. 7. The AS peak point is the apex of the first peak of the acceleration signal, the AO point is the apex of the main peak of the acceleration signal, and the MO is the valley of the secondary peak of the acceleration signal. The AS point in the acceleration signal corresponds to the beginning of atrial contraction, when the inner wall of the chest cavity moves away from the radar generating a negative acceleration. The AO point corresponds to the opening of the aortic valve, due to which the left ventricle starts to contract, the rapid twisting and movement towards the inner wall makes the acceleration decrease. The MO point corresponds to the opening of the mitral valve, where the ventricle starts to fill up with blood, and the wall of the heart moves outward and the acceleration increases [33]. The peak detection algorithm [34] was applied to search and label the AS, AO, and MO peak points.

According to Fig. 7, we know that the acceleration change between AS and AO corresponds to the electrical activity of QRS area, between AO and MO corresponds to the electrical activity of STT area, and between MO and AS corresponds to the electrical activity of PPR area. The reconstruction for each partition is the reconstruction of the ECG signal characteristics. Finally, the acceleration signal and ECG signal are synchronously segmented according to the position of the peak point, and the acceleration signal segment and ECG signal segment datasets are constructed.

2.3 Reconstruction of ECG Signals

Considering that the mechanical and electrical activities of the heart belong to different domain mappings of homologous information [18], we assumed that they share the same hidden features and proceeded to design a cross-domain transformation network for reconstructing the waveform from the acceleration waveform of the heart's motion into the waveform of the ECG signal. End-to-end mapping from RF input to ECG signal output is achieved.

To deal with the complex nonlinear mapping problem between acceleration signals and ECG signals, we use an improved encoder-decoder framework based on sequence-to-sequence (Seq2Seq) learning [35]. Figure 8 shows the detailed architecture of the SS-S2SA network. Specifically, the SS-S2SA network consists of three main parts: the spatio-temporal feature encoder, the ECG reconstruction decoder, and the attention module. The input of the network is acceleration waveform time series data, and the output is ECG signal waveform time series data. During the model training process, the network learns the mapping functions from acceleration signals to hidden features of cardiac motion and from hidden features to ECG signals, respectively. A detailed description of each module in the SS-S2SA network follows.

(1) Spatio-temporal feature encoder, we propose a hybrid CNN-LSTM encoder architecture to extract the spatiotemporal features of the acceleration waveform of each segment of cardiac motion, as shown in Figure 9. Since the acceleration signal is a one-dimensional (1-D) time series, a 1-D CNN network is designed to extract the spatial features. The 1-D CNN proposed in this paper consists of two convolution layers, Conv1 and Conv2, and Pooling1 and Pooling2 two pooling layers, the kernel size of Conv1 is 32×1; the kernel size of Conv2 is 16×1. The kernel size of Conv2 is half of that of Conv1 to ensure that the 1-D CNN is able to learn the spatial features of acceleration waveforms on the basis of different scales. Two maximal pooling layers, Pooling1 and Pooling2, using a pool size of 2, are applied to the feature mapping after each convolutional layer, reducing the size of the extracted features by half to reduce the computational complexity of the subsequent network, retaining

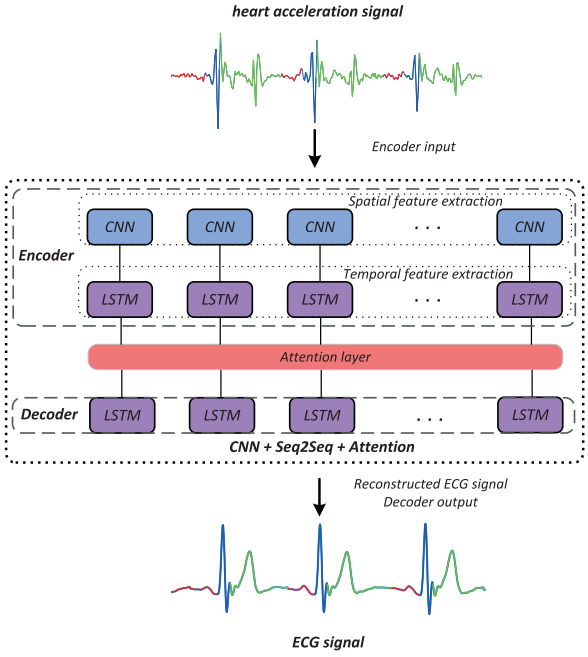


Fig. 8 SS-S2SA network.

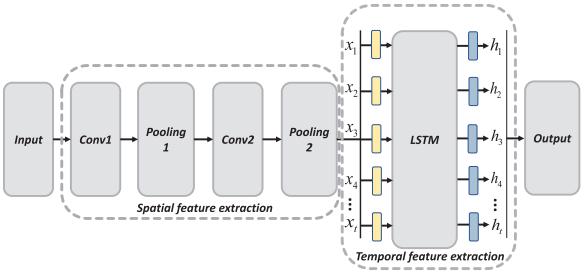


Fig. 9 Spatio-temporal feature encoder.

the main features, and avoiding model overfitting.

After pooling, two LSTM layers are connected, because the acceleration signal has strong temporal features, the correlation vectors are extracted and constructed into a time series by CNN as the input data for LSTM, and the high-dimensional temporal features are extracted from the disordered time series information using the two-layer LSTM. The LSTM adopts forward propagation to capture the temporal features of the data in the forward direction. the basic structure of the LSTM network is shown in Fig. 10 shows. For the input feature sequence $X = (x_1, x_2, \ldots, x_t)$, the output H_t of the forward layer at the moment t is computed as follows:

$$H_t = f\left(WX_t + W'H_{t-1}\right) \tag{15}$$

The final output of the LSTM network Y_t is:

$$Y_t = g\left(UH_t\right) \tag{16}$$

Where, W, W' and U are the weight matrices, obtained through a learning process, and $f(.)$ and $g(.)$ are the activation functions.

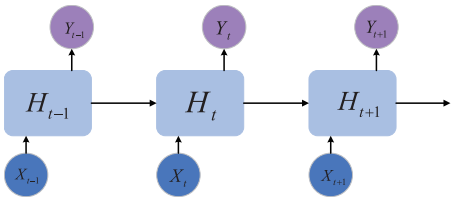


Fig. 10 Illustration of the LSTM network.

(2) Attention module, we provide an attention mechanism for the Seq2Seq model, the core idea of which is to introduce attention weights on the input sequences for weighted decoding after encoding and extraction, and weighted summation of the current moment alignment probability vectors to obtain the contextual weight variables, which accurately focuses the attention information on the main features of the ECG signals, so that the network focuses its attention on the decoder output of the ECG waveforms of the main features to converge faster for better decoding performance. The attention mechanism provided is shown in Fig. 11, and the weight allocation algorithm flows: let the encoder layer input be $[x_1, x_2, \ldots, x_t]$, the output be $[h_1, h_2, \ldots, h_t]$, the implicit layer output at the last moment be y_0, and calculate the similarity weight matrix according to the following equation:

$$\tilde{\alpha}_i = score\left(h_i, y_{i-1}\right) = v^T \tanh\left(w\left[h_i, y_{i-1}\right]\right) \tag{17}$$

Where w, v^T are the weight parameters of the network to be trained. Then $Soft\max$ normalization is performed to make them sum to 1:

$$[\alpha_1, \ldots, \alpha_i] = Soft\max\left([\tilde{\alpha}_1, \ldots, \tilde{\alpha}_i]\right) \tag{18}$$

Weighted summation of h_i with input α_i computes the attentional focus of the input sequence c_i:

$$c_i = \sum_i \alpha_i h_i \tag{19}$$

(3) A decoder module for mapping acceleration waveform data describing the mechanical activity of the heart end-to-end to ECG signal waveform data describing the electrical activity of the heart. Let the decoder layer input be the real ECG signal $[\tilde{y}_1, \tilde{y}_2, \ldots, \tilde{y}_t]$ and the output be $[y_1, y_2, \ldots, y_t]$, the focus of attention of the input sequence c_i is known, $\tilde{y}_i, y_{i-1}, c_i$ is fed into the decoder as an input, is then mapped to the final ECG signal y_i by a linear transformation and an activation function:

$$y_i = \tanh\left(A\left[\tilde{y}_i, y_{i-1}, c_i\right] + b\right) \tag{20}$$

Where A, b is the weight parameter of the network to be trained. By assigning continuously updated weights, the network is made to focus its attention on certain parts of the encoded output rather than relying on the content vector alone, which results in a significant improvement in sequence-to-sequence learning. Finally, the three reconstructed regions of the ECG signal are spliced in chronological order to obtain the complete ECG signal.

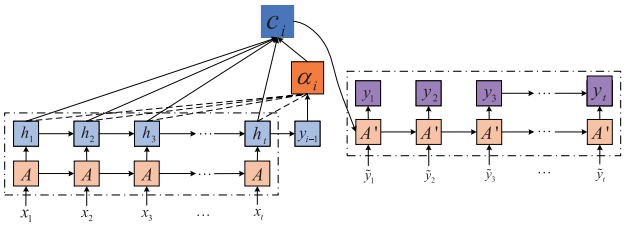


Fig. 11 Decoding mechanisms for incorporating attentional information.

In our proposed SS-S2SA network, the input cardiac motion acceleration data sequence $x = (x_1, x_2, \ldots, x_t)$ and the true ECG data sequence $\tilde{y} = (\tilde{y}_1, \tilde{y}_2, \ldots, \tilde{y}_t)$ are assumed. The input sequence is mapped to the hidden layer representation $[h_1, h_2, \ldots, h_t]$ by an encoder and the predicted ECG data sequence $y = (y_1, y_2, \ldots, y_t)$ is generated using this hidden layer representation. In this process, the absolute error between the predicted and true values is computed using the absolute value loss ($L1$ loss) function:

$$L1 = |\tilde{y} - y| \tag{21}$$

Mean absolute error MAE for all sample values:

$$MAE = \frac{1}{N}\sum_{i=1}^{N}|\tilde{y}_i - y_i| \tag{22}$$

Where N denotes the sequence length. The difference between the true sequence and the predicted sequence is converted to a scalar value by the $L1$ loss function, and reducing this value improves the performance of the model.

3. Experimental Configuration

3.1 Experimental Parameters

Experiments were conducted using an IWR1843 FMCW millimeter-wave radar and a DCA1000 high-speed data acquisition card (Texas Instruments Incorporated) to collect cardiac mechanical activity information. The radar operated in the 77 GHz to 81 GHz band. One transmitting antenna and four receiving antennas were set up. The sampling time was set to 50 us, and the ADC sampling point was 256, so that more subtle cardiac motion information could be captured for each sampling of the echo signal, and the frame frequency was set to 100 Hz, so that the FMCW radar could capture the high-frequency information generated in the process of cardiac motion. The experimental parameters are shown in Table 1.

At the same time, we use TI-ADS1292R to collect the cardiac electrical activity information, the device sampling rate is 500 SPS. Unwanted signals and power line noise are removed from the ECG data by a FIR trap filter with a Hamming Window cutoff frequency of 150 Hz, a trap frequency of 50 Hz, a stopband attenuation of more than 30 dB, and a sampling frequency of 500 samples/s. The FIR trap filter is used to further remove high-frequency noise to obtain a more pure and realistic ECG signal. A low-pass filter with a cutoff frequency of 30 Hz is then used to further remove the high-frequency noise to obtain a more pure and realistic ECG signal.

Table 1 FMCW radar configuration parameters.

Parameter Name	Parameter Value
Start Frequency	77 GHz
Bandwidth	3.9 GHz
Idle Time	10 us
Ramp End Time	60 us
Sample Rate	5 MHz
Frame Periodicity	10 ms
ADC Samples	256
Number of Chirp Loops	2
Number of Frames	3000
Total Time	30s

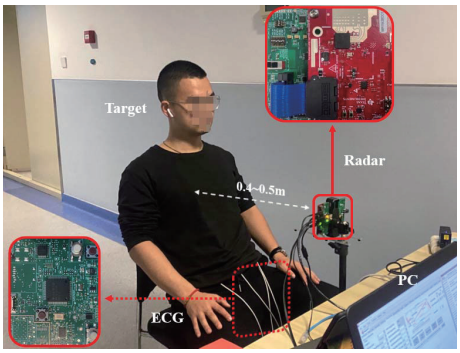


Fig. 12 Experimental environment.

In order to assess the effectiveness of the method proposed in this paper, 10 participants (6 males and 4 females) aged between 23 and 30 years without any heart disease were recruited for this experiment. All volunteers were asked to sit comfortably on a chair 50 cm away from them as shown in Fig. 12. The millimeter wave radar was placed in front of the subject, aiming at the sternum. An ECG sensor was connected to the chest area of the volunteers to collect reference ECG signals. A total of more than 100 experiments were conducted in this study, each for 30 s. The volunteers' heartbeats were approximately 55–110 beats per minute. In total, this paper collected about 6000+ heartbeats. Since long time series data was collected, segmentation of it enables better training of the model. We consider the average time for a healthy subject's heart to beat once to be about 0.8 seconds, and in this paper we take 0.9 seconds as the duration of a heartbeat. Then after applying a peak point based signal segmentation algorithm on these heartbeats, about 18,000 acceleration signal segments were obtained to create the dataset, and finally, the input data was normalized before training to provide more stable training and faster convergence.

3.2 Deep Learning Network Parameters

This experiment trains and saves the best model in lntel(R) Core(TMi7-10700K processor + NVIDIA GeForce RTX 2080 Ti + Pytorch1.10 environment. The deep learning

Table 2 SS-S2SA network parameters.

Parameter Name	Parameter Value
Learning Rate	0.001
Optimizer	Adam
Hidden Size	128
Batch Size	64
Number of Epochs	200

network parameters are shown in Table 2.

4. Discussion on Experimental Methods and Results

To visualize the model performance, we calculated the timing accuracy and morphological accuracy of the reconstruction results separately. The correlation coefficient between the reconstructed ECG and the real ECG was first calculated ρ:

$$\rho = \frac{\sum_{m=1}^{M}\left(y_m - \bar{y}\right)\left(\hat{y}_m - \bar{\hat{y}}\right)}{\sqrt{\sum_{m=1}^{M}\left(y_m - \bar{y}\right)^2}\sqrt{\sum_{m=1}^{M}\left(\hat{y}_m - \bar{\hat{y}}\right)^2}} \tag{23}$$

and calculated as an evaluation metric, where M is the number of all sample points, y_m and $\hat{y}_m$ are the estimated and measured values of the m-th point of the ECG signal, respectively. Moreover, $\bar{y}$ and $\bar{\hat{y}}$ are their mean values. In addition to the correlation coefficients, for a better evaluation of the proposed method, we chose the Pan-Tompkins algorithm to accurately detect R-peaks [34]. Considering that P-peaks usually appear within 0.3 s before R-peaks, we detected the largest peaks appearing within 0.3 s before R-peaks as P-peaks, and detected T-peaks as the largest peaks appearing after R-peaks. As a performance metric, we calculated the mean absolute error (MAE) between the time these peaks appeared on the reconstructed and actual ECG signals. Precisely, for each peak, the MAE was calculated as

$$MAE = \frac{1}{N}\sum_{n=1}^{N}\left|t_{pred}(n) - t_{true}(n)\right| \tag{24}$$

Where N denotes the number of the collected ECG signals, n means the n th ECG signal, and t_{pred} and t_{true} denote the pre-dicted and ground truth timings, respectively. According to the recommendations of the Association for Medical instruments, starts, peaks, or offsets are considered to be correctly detected if they deviate from the true value within 150 ms [36]. In this paper, a better reconstruction result should be one in which the deviation between the actual true value and the predicted value is as small as possible and the correlation coefficient is as high as possible.

4.1 Overall Performance

In this experiment, three subjects were randomly evaluated in the experiment, during which the subjects sat at about 0.5 m from the radar equipment for independent testing, and the reconstruction results were evaluated by applying the acceleration signal segmentation algorithm based on the peak point

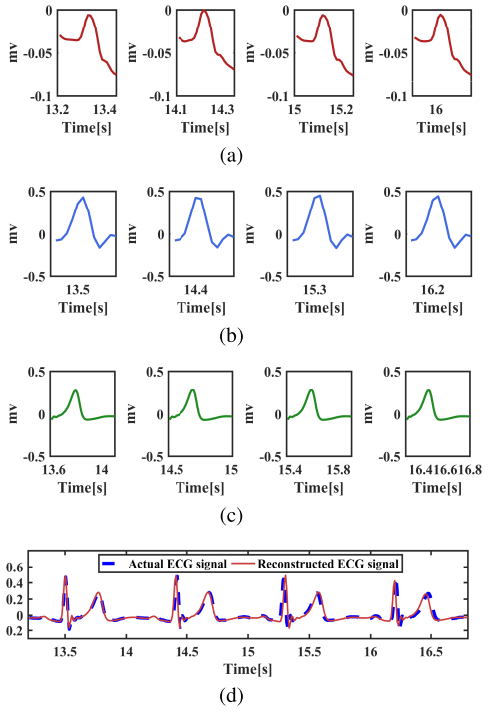


Fig. 13 Reconstruction results. (a) PPR region and (b) QRS region and (c) STT region and (d) the result after splicing.

obtained from the signal segmentation algorithm through the SS-S2SA network, respectively. Figure 13 shows the reconstruction results of one of the subjects between 13.2 and 16.8 s. Where the x (horizontal)-axis represents time and the y (vertical)-axis represents amplitude. Figure 13(a) shows the reconstruction results of the PPR region. Figure 13(b) is the reconstruction result of QRS region. Figure 13(c) is the reconstruction result of STT region. The red solid line in Fig. 13(d) is the complete ECG signal obtained by splicing the reconstruction results of the three regions in (a)(b)(c) in chronological order. It can be seen that the SS-S2SA method proposed in this paper is able to accurately reconstruct the individual bands and then splice to obtain the complete ECG signal.

Table 3 shows the final results of this experiment, where MAE is calculated as the mean absolute error of P-peaks, Q-peaks, R-peaks, S-peaks, and T-peaks timings in the ECG signal sequence, and the MAEs of the test set are within the MAE range allowed by the international standard (150 ms). Corr is denoted as the correlation coefficient of the reconstructed signal and the real signal. We improve the learning ability of non-significant locations by introducing partitioned reconstruction, and at the same time make the SS-S2SA network focus on learning the mapping relation of peak locations through the attention mechanism. The results confirm that the method proposed in this paper can accurately reconstruct ECG signals. From Table 3, it can be seen that the timing of the individual peaks of the ECG signal was accurately estimated, and the timed MAE of the P-peak, the R-peak, and the T-peak were 13.9 ms, 8.1 ms, and 11.1 ms,

Table 3 The results of the assessment of different test subjects.

	MAE(ms)			Corr.
	P	R	T	
Target 1	12.9	6.3	11.0	0.93
Target 2	15.2	8.7	10.5	0.91
Target 3	13.5	9.3	11.8	0.91
Ave.	13.9	8.1	11.1	0.92

Table 4 Results for different networks and different hyperparameter values.

Configuration	MAE(ms)			Corr.
	P	R	T	
Seq2Seq(num-layers=8, hidden-size=128)	22.7	14.5	19.3	0.86
SS-S2SA(num-layers=4, hidden-size=128)	15.3	9.1	14.2	0.89
SS-S2SA(num-layers=8, hidden-size=128)	12.9	6.3	11.0	0.93
SS-S2SA(num-layers=8, hidden-size=256)	14.0	7.8	12.6	0.91

respectively, with an average correlation coefficient of 0.92.

4.2 Different Networks and Different Hyperparameters

This experiment provides insight into the potential impact on the sequence-to-sequence mapping task by observing the differences in ECG signal reconstruction performance under different neural networks and different hyperparameters. First, comparison experiments were conducted using a traditional sequence-to-sequence network (Seq2Seq) and the SS-S2SA (Seq2Seq+Attention) network proposed in this paper, respectively. Then, different hyper-parameter settings of SS-S2SA network are used, and the number of coding-decoding layers are all 4 and 8 respectively. The number of hidden layers are 128 and 256 respectively. The experimental results are shown in Table 4.

In Table 4, the above values were obtained by calculating the average of the reconstruction results for individual segments of the test set. The timing MAE results of P-peak, R-peak, and T-peak are all within tolerance in both Seq2Seq and SS-S2SA networks. And the peak timing MAE of the SS-S2SA model is significantly lower than that of the Seq2Seq model, which indicates that the SS-S2SA network predicts the peak location more accurately than the Seq2Seq network. This is because the hybrid CNN-LSTM encoder structure is used in the SS-S2SA network proposed in this paper to extract the temporal and spatial features of cardiac motion acceleration waveforms, and the attention information is introduced during the decoding process so that the network focuses its attention on the main features of the ECG waveforms output from the decoder to converge faster to obtain better decoding performance. In addition, under the same hyperparameters (num-layers=8, hidden-size=128), the correlation coefficient of the Seq2Seq model is 0.86. The correlation coefficient of the SS-S2SA model is 0.93. This indicates that the SS-S2SA network is significantly better than the Seq2Seq network in terms of correlation coefficient. When the num-layers is 8 and the hidden-size is 128, the correlation coefficient between the predicted ECG signal and the real ECG signal is larger (0.93), and the timing MAE of the P, R, and T peaks is smaller, which are 12.9 ms, 6.3 ms, and 11.0 ms, respectively. In summary, the SS-S2SA network is more suitable for the reconstruction of ECG signals.

Table 5 Subject details.

	Age	Sex	Height(cm)	Weight(kg)	BMI
Target 1	26	M	172	63	21.3
Target 2	25	F	168	60	22.0
Target 3	22	F	164	46	17.1
Target 4	25	M	182	92	27.8
Target 5	28	M	175	89	29.1

4.3 Assessment of Different Populations

This experiment aims to investigate the effects of different population characteristics (e.g., height, weight, and gender) on the experimental results. By observing the differences in ECG signal reconstruction performance between different groups of people, such as tall, short, fat, thin, male and female, we aim to gain insights into the potential effects of these factors on the experimental results, thus providing a more comprehensive perspective for the study in this paper.

Therefore, under the premise of ensuring that the age distribution is basically the same, we selected 5 subjects with significant population characteristics (thin, normal, overweight, obese) and a balanced distribution of men and women (3 males and 2 females) from healthy subjects for evaluation experiments. The BMI index (i.e., Body Mass Index, BMI) of each subject is then calculated from height and weight, as shown in Table 5, where M denotes male, F denotes female. The four fat and thin conditions indicated by BMI are: thin (<=18.4), normal (18.5–23.9), overweight (24.0–27.9), and obese (28.0–32).

In our experiments, we performed non-contact ECG signal reconstruction on the above subjects, and the reconstruction performance is shown in Fig. 15(a) and (b). In terms of reconstruction performance, the male subjects are better than the female subjects when the BMI indexes are similar, which is shown by the low timed MAE and high correlation coefficients for the P-peak, R-peak, and T-peak of the ECG signals. This is because in general, male hearts are larger and stronger than female hearts, and male hearts usually have larger myocardial mass and higher cardiac output, so the beating force of male hearts may be slightly stronger, which improves the strength of the heart motion acceleration signals in the radar echoes, as shown in Fig. 14(a) and (b), making it easier to reconstruct the cardiac electrical signals.

In addition, the reconstruction performance of obese subjects is lower than that of normal or thin subjects because the electromagnetic waves emitted by the radar undergo absorption and scattering phenomena when traversing the body fat. Therefore, for people with more fat, the electromagnetic wave will experience more absorption and scattering processes when traversing the body fat, resulting in a weakening of the cardiac motion echo signal strength, as shown in Fig. 14(c) and (d), which degrades the subsequent reconstruction performance. So far, variability in subjects' BMI and cardiac waveforms have been responsible for the differ-

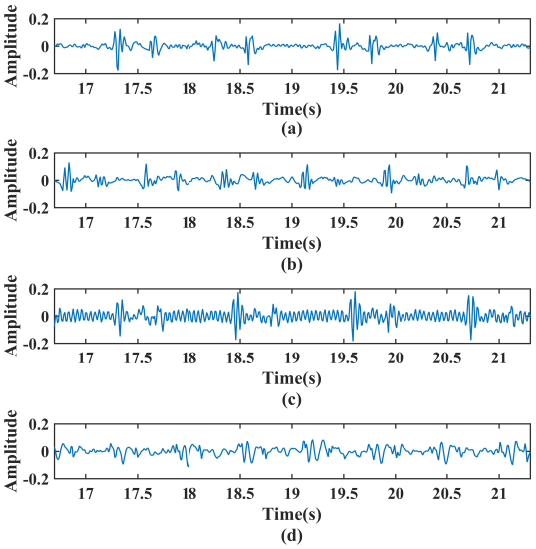


Fig. 14 Cardiac motion acceleration signals in (a) male subjects and (b) female subjects and (c) subjects with a BMI of 21.3 and (d) subjects with a BMI of 29.1.

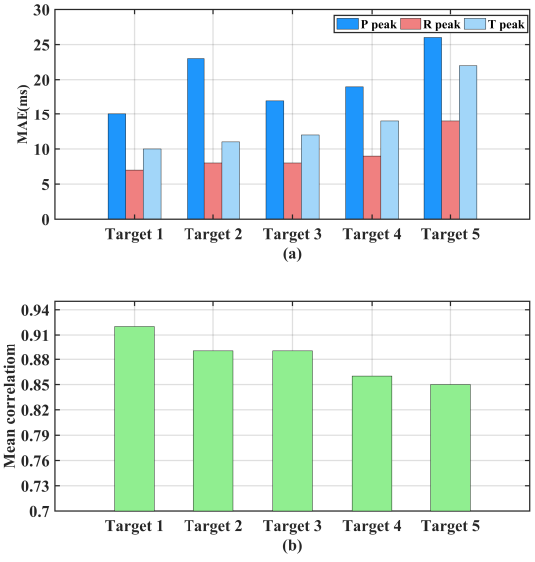


Fig. 15 Results for targets at 0.5 m. The (a) MAE and (b) mean correlation.

ences in estimation accuracy between subjects. In addition, exploring the effects of blood pressure, emotion, and other factors on ECG reconstruction performance will remain a part of future research.

5. Conclusion

In this paper, an ECG signal reconstruction method based on FMCW radar and SS-S2SA network is proposed, which includes two parts: signal preprocessing and reconstruction of ECG signals. In the signal preprocessing part, this paper proposes an improved DACM algorithm and multi-neighborhood differentiator, which extracts the cardiac motion acceleration information in the range of 0.9–50 Hz from the RF signals as the input to the network. In the part of reconstructing the ECG signal, this paper proposes a combination of signal processing and deep learning network, firstly, applying a segmentation algorithm to synchronize the segmentation of acceleration signal and ECG signal, and then mapping the cardiac motion acceleration signal to the ECG signal by SS-S2SA network. The experimental results show that the SS-S2SA network has a higher reconstruction performance, which is manifested by the fact that the timing MAEs of the P-peak, R-peak, and T-peak of the predicted ECG signals are within the required range, and the average correlation coefficient of the predicted signals with the real signals is about 0.92. In the future work, it is necessary to further improve the reconstruction performance in accordance with the accuracy requirements in medical applications. In addition, the inclusion of research and evaluation of people suffering from heart disease will also be a future research topic.

Acknowledgments

This work was supported by Jilin Province Science and Technology Development Plan Project under Grant No.YDZJ202201ZYTS511. The authors are sincerely grateful for their support.

References

[1] G.A. Roth, C.O. Johnson, K.H. Abate, F. Abd-Allah, M. Ahmed, K. Alam, T. Alam, N. Alvis-Guzman, H. Ansari, J. Ärnlöv, et al., "The burden of cardiovascular diseases among US states, 1990–2016," JAMA Cardiol., vol.3, no.5, pp.375–389, 2018.

[2] E. Wilkins, L. Wilson, K. Wickramasinghe, P. Bhatnagar, J. Leal, R. Luengo-Fernandez, R. Burns, M. Rayner, and N. Townsend, "European cardiovascular disease statistics 2017," European Heart Network, 2017.

[3] S. Yasuda, Y. Miyamoto, and H. Ogawa, "Current status of cardiovascular medicine in the aging society of japan," Circulation, vol.138, no.10, pp.965–967, 2018.

[4] M.F.u. Hassan, D. Lai, and Y. Bu, "Characterization of single lead continuous ECG recording with various dry electrodes," Proc. 2019 3rd International Conference on Computational Biology and Bioinformatics, Nagoya, Japan, pp.76–79, Oct. 2019.

[5] S. Preejith, R. Dhinesh, J. Joseph, and M. Sivaprakasam, "Wearable ECG platform for continuous cardiac monitoring," 2016 38th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC), Norlando, FL, USA, pp.623–626, IEEE, 2016.

[6] D. Azariadi, V. Tsoutsouras, S. Xydis, and D. Soudris, "ECG signal analysis and arrhythmia detection on iot wearable medical devices," 2016 5th International conference on modern circuits and systems technologies (MOCAST), Thessaloniki, Greece, pp.1–4, IEEE, 2016.

[7] Y. Hu, X. Peng, and T. Toda, "A novel remote-tracking heart rate measurement method based on stepping motor and mm-Wave FMCW radar," IEICE Trans. Commun., vol.E107-B, no.6, pp.470–486, June 2024.

[8] M. Alizadeh, G. Shaker, J.C.M. De Almeida, P.P. Morita, and S. Safavi-Naeini, "Remote monitoring of human vital signs using mm-Wave FMCW radar," IEEE Access, vol.7, pp.54958–54968, 2019.

[9] T.K.V. Dai, K. Oleksak, T. Kvelashvili, F. Foroughian, C. Bauder,

P. Theilmann, A.E. Fathy, and O. Kilic, "Enhancement of remote vital sign monitoring detection accuracy using multiple-input multiple-output 77 GHz FMCW radar," IEEE J. Electromagn., RF Microw. Med. Biol., vol.6, no.1, pp.111–122, March 2022.

[10] K. Gupta, M. Srinivas, J. Soumya, O.J. Pandey, and L.R. Cenkeramaddi, "Automatic contact-less monitoring of breathing rate and heart rate utilizing the fusion of mmWave radar and camera steering system," IEEE Sensors J., vol.22, no.22, pp.22179–22191, Nov. 2022.

[11] Y. Hu and T. Toda, "The effect of multi-directional on remote heart rate measurement using PA-LI joint ICEEMDAN method with mm-Wave FMCW radar," IEICE Trans. Commun., vol.E105-B, no.2, pp.159–167, Feb. 2022.

[12] C. Zhu, M. Balle, B. Zhang, Q. Lv, Y. Li, Z. Zhu, X. Wang, X. Li, and L. Ran, "Doppler cardiogram detected by a V-band doppler radar sensor," IEEE Trans. Microw. Theory Techn., vol.70, no.1, pp.521–531, Jan. 2022.

[13] W. Xia, Y. Li, and S. Dong, "Radar-based high-accuracy cardiac activity sensing," IEEE Trans. Instrum. Meas., vol.70, pp.1–13, Jan. 2021.

[14] C. Xu, H. Li, Z. Li, H. Zhang, A.S. Rathore, X. Chen, K. Wang, M.c. Huang, and W. Xu, "CardiacWave: A mmWave-based scheme of non-contact and high-definition heart activity computing," Proc. ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies, vol.5, no.3, pp.1–26, 2021.

[15] U. Ha, S. Assana, and F. Adib, "Contactless seismocardiography via deep learning radars," Proc. 26th Annual International Conference on Mobile Computing and Networking, New York, USA, pp.1–14, 2020.

[16] Z. Xia, M.M.H. Shandhi, O.T. Inan, and Y. Zhang, "Non-contact sensing of seismocardiogram signals using microwave doppler radar," IEEE Sensors J., vol.18, no.14, pp.5956–5964, July 2018.

[17] Z. Xia, M.M.H. Shandhi, Y. Li, O.T. Inan, and Y. Zhang, "The delineation of fiducial points for non-contact radar seismocardiogram signals without concurrent ECG," IEEE J. Biomed. Health Inform., vol.25, no.4, pp.1031–1040, April 2021.

[18] I. Odinaka, J.A. O'Sullivan, E.J. Sirevaag, and J.W. Rohrbaugh, "Cardiovascular biometrics: Combining mechanical and electrical signals," IEEE Trans. Inf. Forensics Security, vol.10, no.1, pp.16–27, Jan. 2015.

[19] W.G. Buratto, R.N. Muniz, A. Nied, and G.V. Gonzalez, "Seq2Seq-LSTM with attention for electricity load forecasting in brazil," IEEE Access, vol.12, pp.30020–30029, 2024.

[20] J. Park, H. Cho, W. Hwang, R.K. Balan, and J. Ko, "Deep ECG wave estimation model with seismograph sensor (poster)," Proc. 17th Annual International Conference on Mobile Systems, Applications, and Services, pp.568–569, June 2019.

[21] Y. Li, Z. Xia, and Y. Zhang, "Standalone systolic profile detection of non-contact SCG signal with LSTM network," IEEE Sensors J., vol.20, no.6, pp.3123–3131, Dec. 2019.

[22] K. Yamamoto and T. Ohtsuki, "Non-contact heartbeat detection by heartbeat signal reconstruction based on spectrogram analysis with convolutional LSTM," IEEE Access, vol.8, pp.123603–123613, 2020.

[23] K. Yamamoto, R. Hiromatsu, and T. Ohtsuki, "ECG signal reconstruction via doppler sensor by hybrid deep learning model with CNN and LSTM," IEEE Access, vol.8, pp.130551–130560, July 2020.

[24] D. Toda, R. Anzai, K. Ichige, R. Saito, and D. Ueki, "ECG signal reconstruction using FMCW radar and convolutional neural network," 2021 20th International Symposium on Communications and Information Technologies (ISCIT), pp.176–181, IEEE, Oct. 2021.

[25] J. Chen, D. Zhang, Z. Wu, F. Zhou, Q. Sun, and Y. Chen, "Contactless electrocardiogram monitoring with millimeter wave radar," IEEE Trans. Mobile Comput., vol.23, no.1, pp.270–285, Jan. 2024.

[26] Z. Wang, C. Li, S. Tu, and Z. Liu, "A novel rotary FMCW radar for omnidirectional multiperson localization and vital signs detection," IEEE Trans. Microw. Theory Techn., vol.72, no.3, pp.1886–1899, March 2024.

[27] M. Mao and X. Han, "Optimization research on radar signal processing technology based on MTI-MTD," 2023 IEEE 3rd International Conference on Data Science and Computer Application (ICDSCA), Dalian, China, pp.480–486, IEEE, 2023.

[28] J. Wang, X. Wang, L. Chen, J. Huangfu, C. Li, and L. Ran, "Non-contact distance and amplitude-independent vibration measurement based on an extended DACM algorithm," IEEE Trans. Instrum. Meas., vol.63, no.1, pp.145–153, Jan. 2014.

[29] W. Xu, C. Gu, and J.F. Mao, "Noncontact high-linear motion sensing based on a modified differentiate and cross-multiply algorithm," 2020 IEEE/MTT-S International Microwave Symposium (IMS), Los Angeles, CA, USA, pp.619–622, IEEE, 2020.

[30] X. Peng, Y. Hu, and T. Toda, "Design and evaluation of remote tracking heart rate measurement based on millimeter-wave FMCW radar," IEICE Commun. Express, vol.11, no.6, pp.330–335, June 2022.

[31] K. Yamamoto, K. Toyoda, and T. Ohtsuki, "Spectrogram-based non-contact RRI estimation by accurate peak detection algorithm," IEEE Access, vol.6, pp.60369–60379, Oct. 2018.

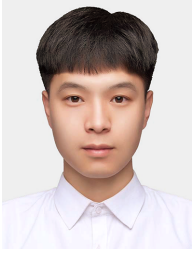
[32] S. Dong, L. Wen, Z. Zhang, C. Gu, and J. Mao, "Contactless measurement of human systolic time intervals based on Doppler cardiograms in clinical environment," IEEE Microw. Wireless Compon. Lett., vol.32, no.6, pp.796–799, June 2022.

[33] S. Sieciński, P.S. Kostka, and E.J. Tkacz, "Gyrocardiography: A review of the definition, history, waveform description, and applications," Sensors, vol.20, no.22, p.6675, 2020.

[34] J. Pan and W.J. Tompkins, "A real-time QRS detection algorithm," IEEE Trans. Biomed. Eng., vol.BME-32, no.3, pp.230–236, March 1985.

[35] D. Coquenet, C. Chatelain, and T. Paquet, "End-to-end handwritten paragraph text recognition using a vertical attention network," IEEE Trans. Pattern Anal. Mach. Intell., vol.45, no.1, pp.508–524, Jan. 2023.

[36] ANSI/AAMI/ISO EC57:1998 (R2008), Association for the Advancement of Medical Instrumentation, 1999.



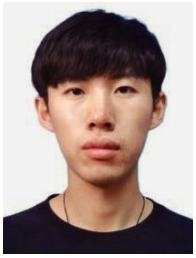
Renwei Cui received the B.Eng. degree in electronic information engineering from the Beihua University, Jilin, China, in 2022, where he is currently pursuing the M.S. degree in Control Science and Engineering from Changchun University of Science and Technology, Jilin, China. His research interests include radar signal processing, biomedical signal processing, and vital sign detection.



Wei Cui received the M.S. and Ph.D. degrees in electronics from the Changchun University of Science and Technology (CUST), Jilin, China, in 2004 and 2010, respec-tively. He is currently associate professor with CUST. His research interests include radar signal processing and intelligent robots.



Yujian Cai received the B.Eng. degree in Electronic Information Engineering from Yanbian University, Jilin, China, in 2021. He is currently pursuing the Ph.D. degree in Information and Communication Engineering with Changchun University of Science and Technology, Jilin, China. His research interests include radar signal processing, emotion recognition and vital sign detection.



Yu Yan received the B.Eng. degree in electronic information engineering from the Changchun University of Technology, Jilin, China, where he is currently pursuing the M.S. degree in Control Engineering from Changchun University of Science and Technology, Jilin, China. His research inter-ests include signal processing, control algorithms and intelligent robots.